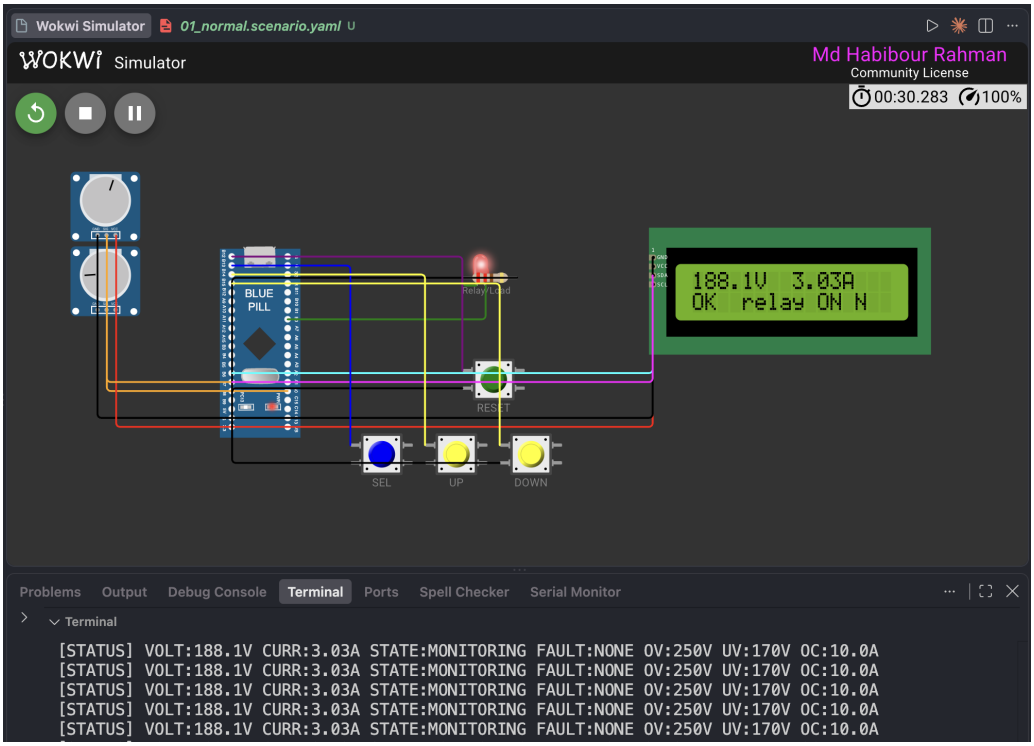
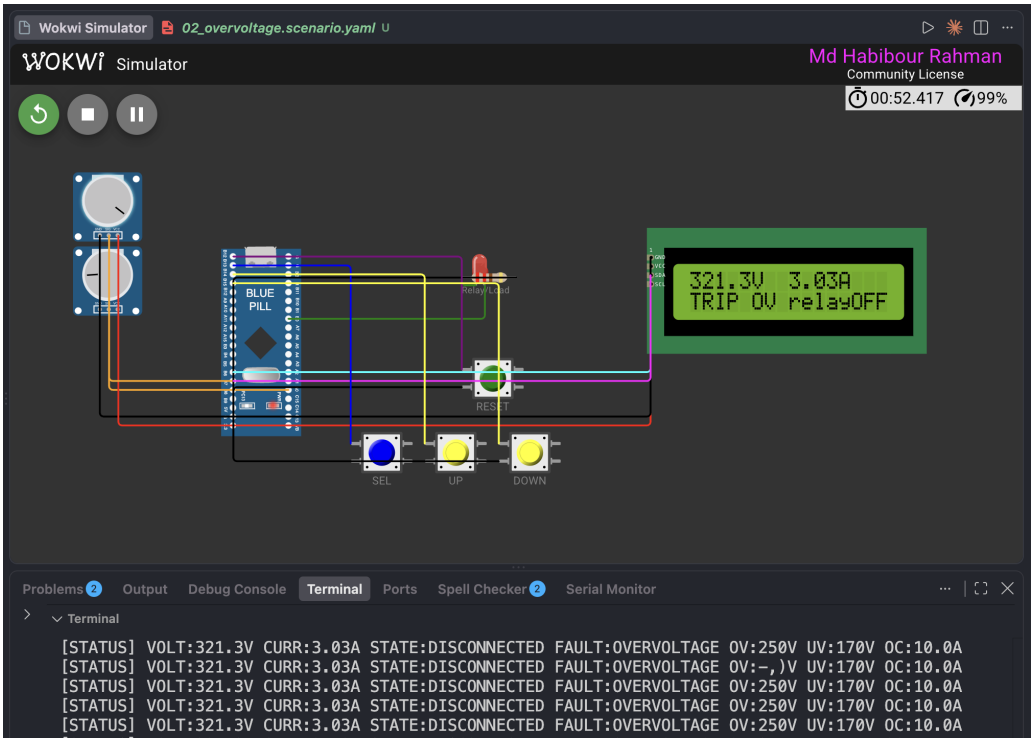
Appendix - Simulation screenshots

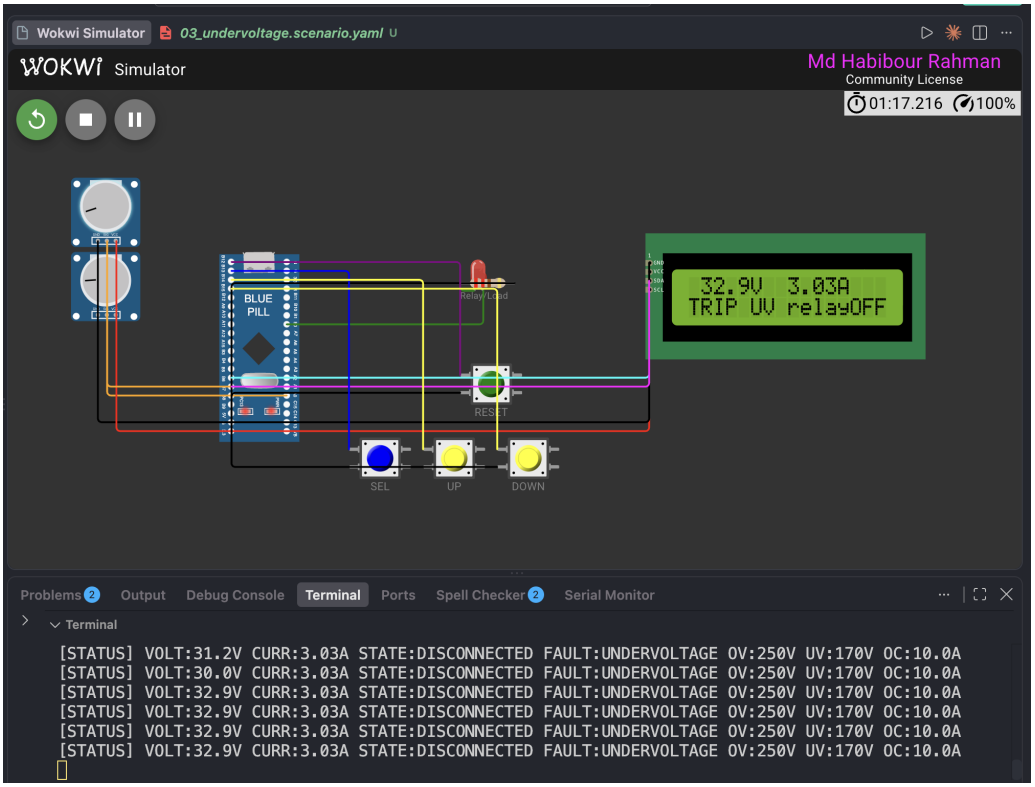
Wokwi simulation of the firmware, one capture per scenario. Potentiometer 1 = voltage (0-330 V), potentiometer 2 = current (0-20 A); the red LED on PB0 is the relay.



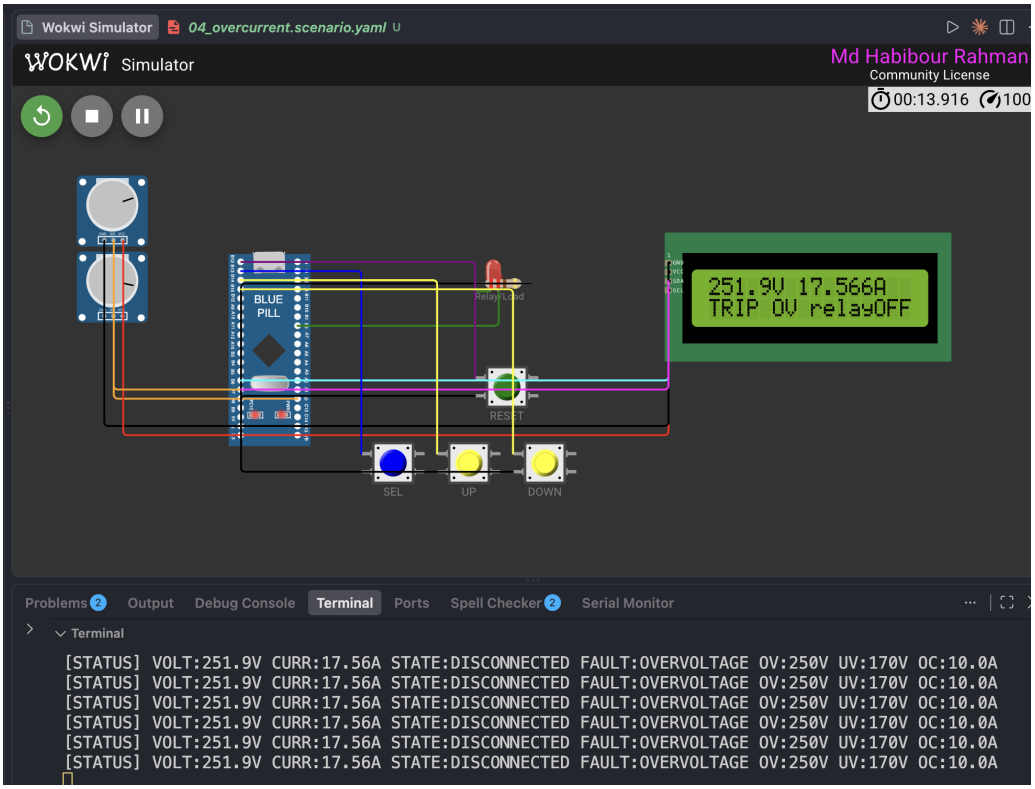
01 Normal operation. 188.1 V, 3.03 A inside the limits. State MONITORING, LCD *OK relay ON*. Relay LED on.



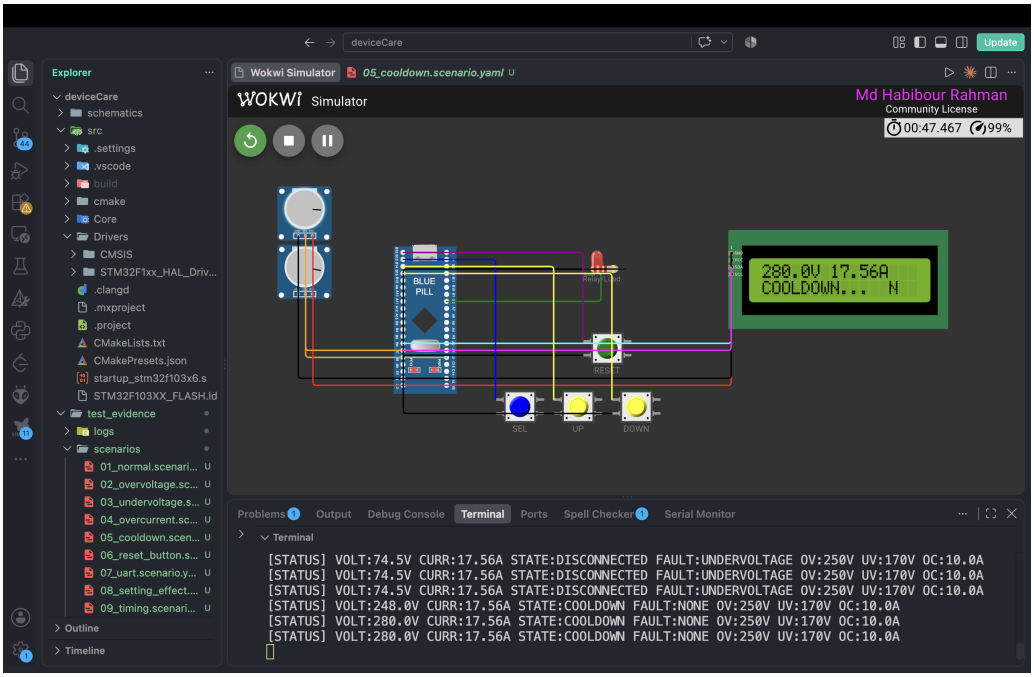
02 Over-voltage. 321.3 V is above the 250 V limit. State DISCONNECTED, LCD *TRIP OV relayOFF*. Relay LED off.



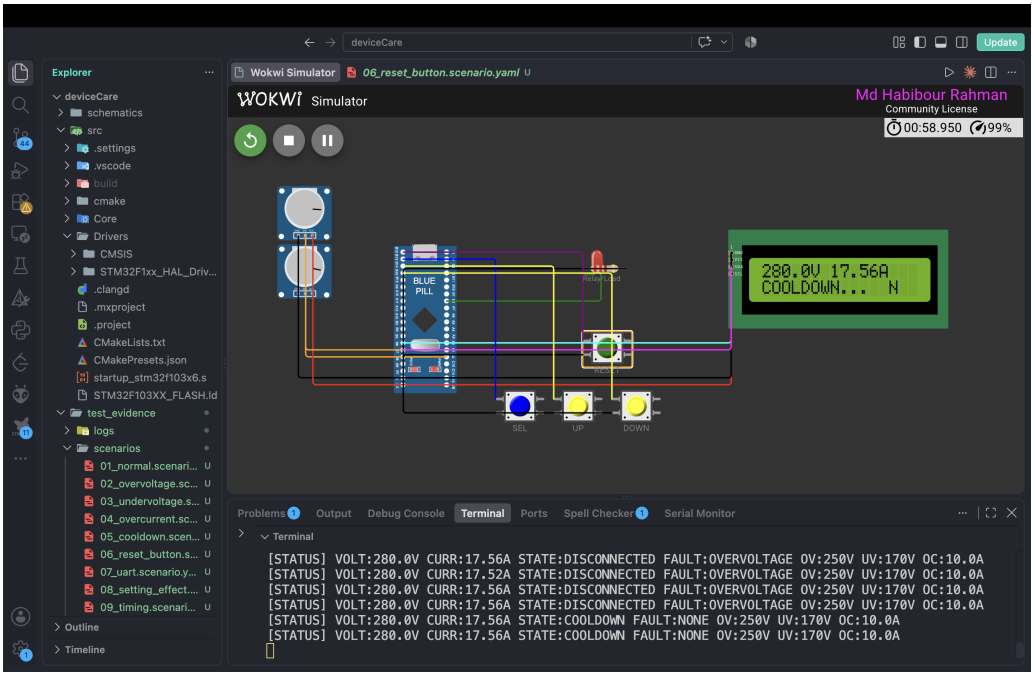
03 Under-voltage. 32.9 V is below the 170 V limit. State DISCONNECTED, LCD *TRIP UV relayOFF*. Relay LED off.



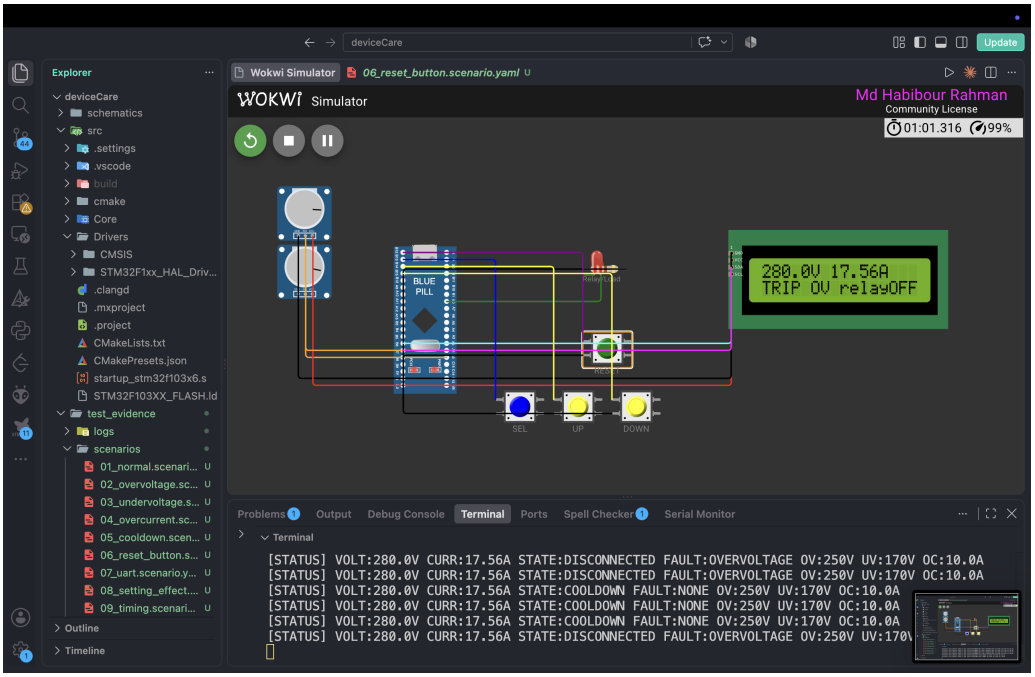
04 Over-current. 17.56 A is above the 10 A limit and the relay is off. Voltage was also 251.9 V, so the fault shown is OV (voltage is checked first); the over-current-only case is in logs/04_overcurrent.log. Relay LED off.



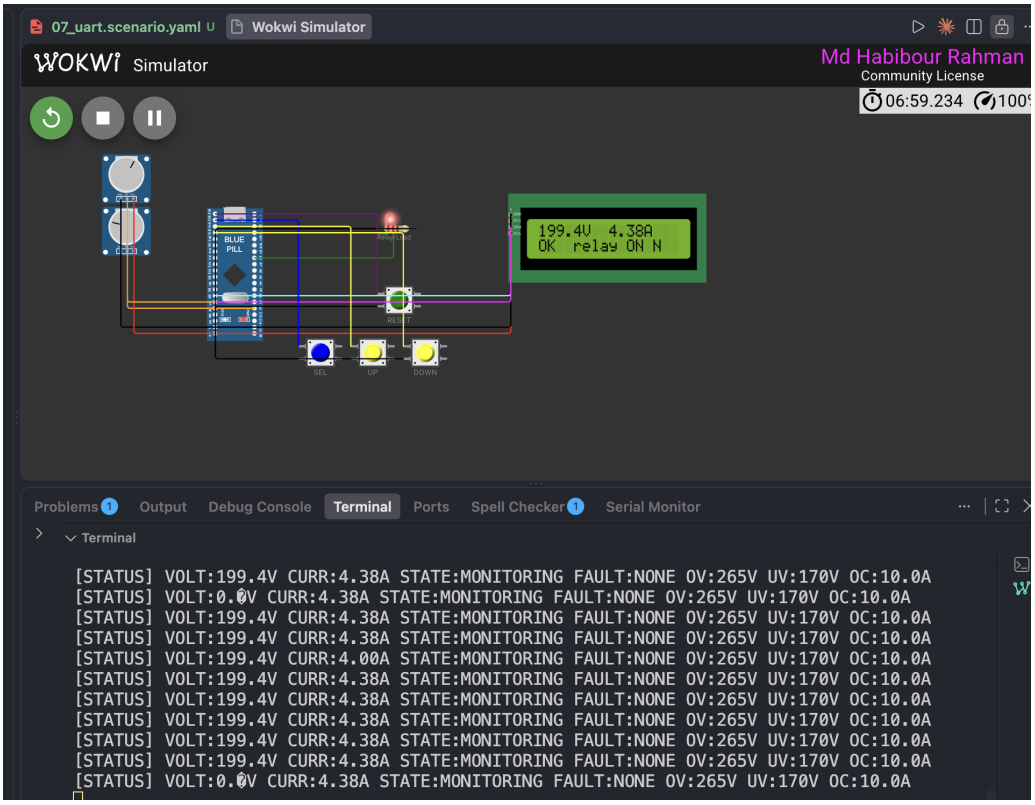
05 Cooldown. After the under-voltage clears the state is COOLDOWN, LCD *COOLDOWN...*. Relay LED stays off for 3 s, then reconnects.



06 Reset button, pressed. With an over-voltage fault active, RESET is pressed and the state goes to COOLDOWN. Relay LED off.



06 Reset button, outcome. 280 V is still applied, so after the cooldown the device trips again: DISCONNECTED / OVERVOLTAGE, LCD *TRIP OV relayOFF*.



07 Limit changed. The status line reports OV:265V instead of the 250 V default, with 199.4 V inside the window: MONITORING, relay on. UART input could not be verified in Wokwi, so this does not show how the limit was set.